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PHAR0002: An Introduction to the Mechanisms of Drug Action.

Will it help to take two ibuprofen?
Welcome to the Hill-Langmuir Equation.

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This handout has been modified from the original text developed by Prof. D.H. Jenkinson who taught at UCL for many years and eventually converted his notes into an excellent chapter in the *Textbook of Receptor Pharmacology*, Ed., Foreman, J.C., and Johansen, T., CRC Press, 1996, 2003, 2010.

Aims:
1) To understand the law of mass action and to be able to apply it to simple reactions at equilibrium.
2) To know what the Hill-Langmuir equation is and to be able to apply it.
3) To understand the assumptions that must be made in relating the Hill-Langmuir equation of receptor occupancy to the magnitude of tissue response.

Receptors and Concentration-Response Relationships

Modelling the relationship between agonist concentration and tissue response

In this lecture we consider a very basic question that lies at the heart of pharmacology; "if I take twice the dose of a drug, will this be twice as effective?". In other words what is the relationship between drug dose and the response of tissues in our body? It is surprisingly difficult to answer this question because the result of doubling a drug dose depends on many factors. These include drug catabolism (how the drug is broken down by metabolic processes), how the drug distributes itself in the body, the concentration-dependence of the drug binding to its site of action and the (often complicated) relationship between drug binding and tissue response. We will therefore begin to examine the relation between dose and response by studying the simplest of these three points - the concentration-dependence of drug binding to its site of action.

The relationship between drug concentration and receptor occupancy

To begin with we will consider the simplest possible model for the combination of an agonist, A, with its receptor, R, (as was originally done when pharmacologists first examined this issue). We might picture something like the scheme below where A combines with R to form an active AR complex:

$$A + R \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} AR$$

In order to be able to analyse the behaviour in this sort of equilibrium we need to know the **law of mass action**. The law of mass action states that *the rate of a reaction is proportional to the concentration of its reactants*. To see how this law works let us consider the forward reaction rate, i.e. the rate at which agonist and receptor combine to form a complex; we consider only half of the scheme above as follows:

$$A + R \xrightarrow{k_{+1}} AR$$

According to the law of mass action;

The rate of formation for the complex between A and R $\propto [A][R]$

Where [A] is the concentration of unbound agonist and [R] is the concentration of receptor which does not have ligand bound. We can take this a step further by inserting a rate constant (that we will call k_{+1}) into the above expression. Then we can dispense with the proportionality sign and equate terms giving:

The rate of formation for the complex between A and R = $k_{+1}[A][R]$ eq. 1

Now we must turn our attention to the rate at that the complex between A and R breaks down. Treating it in a similar fashion we consider the other half of our original scheme:

$$A + R \xleftarrow[k_{-1}]{} AR$$

Again the law of mass action tells us that:

The rate of breakdown for the complex between A and R $\propto [AR]$,

where [AR] is the concentration of receptor which has agonist bound. Again, by inserting a rate constant (which we will call k_{-1}) we can equate these terms giving;

The rate of breakdown for the complex between A and R =$k_{-1}[AR]$ eq. 2

Here the rate constants which we have arbitrarily named k_{+1} and k_{-1} are the *association rate constant* (which has units of $M^{-1}\ s^{-1}$) and the *dissociation rate constant* (which has units of s^{-1}) respectively.

We now apply these equations to a case where the system reaches equilibrium. Because it is an equilibrium condition the quantities of receptor both with and without agonist bound remain unchanged. This can only happen when the rate at which the AR complex is formed is equal to the rate at which it breaks down. In such a situation the right hand side of equations 1 and 2 must be equal to each other giving;

$$k_{+1}[A][R] = k_{-1}[AR]. \qquad \text{eq.3}$$

Before we go further, it may well seem odd to refer to receptor *concentrations* in the context of membrane receptors (the targets for many drugs) because they are free to move only in the plane of the membrane (and then perhaps to no more than a limited extent, since many receptors are anchored). However, the model can be formulated equally well in terms of the proportions of a population of binding sites that are either free or occupied by a drug molecule. Further, it is more useful to think in terms of the proportion of activated receptors because whatever the concentration of activated receptor is, if the *proportion* is very small, for example if 99.9% of the receptor is still inactive (unbound), then this concentration is unlikely to produce any response. So, let us reformulate our equations in terms of proportions. To do this we must first define the total

concentration of receptors. We will call this quantity $[R]_T$. In our simple model the receptor is either free or combined with agonist (there are no other possibilities) so we can write:

$$[R] + [\mathrm{AR}] = [R]_T \qquad \text{eq.4}$$

Now we define p_R as the proportion of free (unbound) receptor as:

$$p_R = \frac{[R]}{[R]_T} \qquad \text{eq.5}$$

Similarly we define the proportion occupied, p_{AR}, as:

$$p_{AR} = \frac{[\mathrm{AR}]}{[R]_T} \qquad \text{eq.6}$$

With these definitions in hand we can now return to equation 3 and divide both sides by $[R]_T$ giving:

$$k_{+1}[\mathrm{A}]p_\mathrm{R} = k_{-1}p_\mathrm{AR} \qquad \text{eq.7}$$

Since we are, for the present, concerned only with equilibrium conditions and not with kinetics, we can combine k_{+1} and k_{-1} to form a new constant, $K_A = k_{-1}/k_{+1}$. K_A is a *dissociation equilibrium constant* (further details in Appendix 1), though often abbreviated to either *equilibrium constant* or *dissociation constant*:

$$[\mathrm{A}]p_\mathrm{R} = K_\mathrm{A}p_\mathrm{AR} \qquad \text{eq.8}$$

Next, by dividing both sides of equation 4 by $[R]_T$ we get:

$$p_\mathrm{R} + p_\mathrm{AR} = 1 \qquad \text{eq.9}$$

Substituting for p_R (from equation 8) into equation 9 gives;

$$\frac{K_\mathrm{A}}{[\mathrm{A}]}p_\mathrm{AR} + p_\mathrm{AR} = 1 \qquad \text{eq.10}$$

Hence by rearrangement we get[1]:

$$p_\mathrm{AR} = \frac{[\mathrm{A}]}{K_\mathrm{A}+[\mathrm{A}]} \qquad \text{eq.11}$$

This is the important ***Hill-Langmuir equation***. A.V. Hill was the first to apply the law of mass action to the relationship between drug concentration and receptor occupancy at equilibrium and to the rate at which this equilibrium is approached. The physical chemist I. Langmuir showed a few years later that a similar equation (the *Langmuir adsorption isotherm)* applies to the adsorption of gases to the surface of a metal. NOTE: Although we have derived this equation by considering an agonist-receptor interaction, it is valid for any simple bimolecular reaction of the type illustrated at the start of these notes.

In deriving Equation 11 you may have wondered why we have not made a clear distinction between the free concentration of agonist [A] and the total concentration $[A]_T$ since we did this for the receptor population. The reason is that, in practice, we always apply this equation to cases where [A] does not appreciably change as drug receptor complexes are formed. In effect, the drug is considered to be present in such excess that it is scarcely depleted by combination of a little of it with the receptors. Thus, the free agonist concentration [A] is known, since $[A]_T$ =[A] + [AR] and under this circumstance $[A]_T \approx$[A].

The relationship between p_{AR} and [A] predicted by Equation 11 is illustrated in **Figure 1**. The concentration of A has been plotted using a linear (left) and a logarithmic (right) scale. (If you are not familiar with logarithms a brief introduction is given in Appendix 3.) The value of K_A has been taken to be 1 μM. (Note that from equation 11 that when [A] = K_A, p_{AR} = 0.5, i.e., half of the receptors are occupied).

With a logarithmic scale, the slope of the line initially increases: the curve has the form of an elongated 'S', and hence is said to be *sigmoidal*. In contrast, with a linear (arithmetic) scale for [A], there is no sigmoidicity: the slope declines as [A] increases and the curve forms part of a rectangular hyperbola.

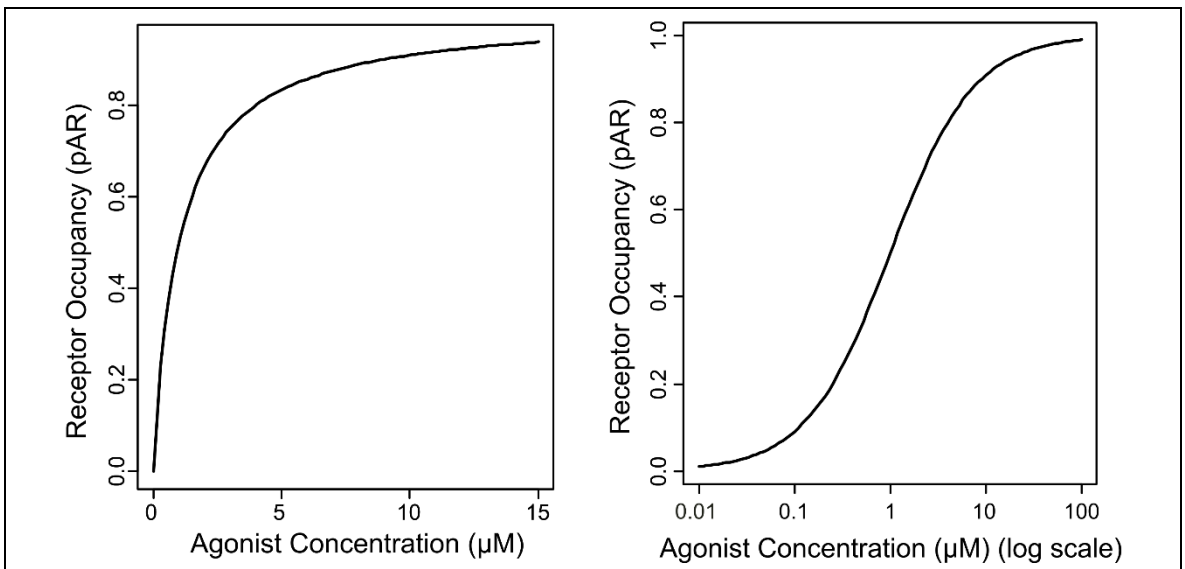


Figure 1. The relationship between receptor occupancy and drug concentration (left, linear scale; right, log scale), as predicted by the Hill-Langmuir equation. K_A has been taken to be 1 μM for both curves.

Equation 11 can be rearranged to:

$$\frac{p_{AR}}{1-p_{AR}} = \frac{[A]}{K_A} \qquad \text{eq.12}$$

Taking logs, we have:

$$\log\left(\frac{p_{AR}}{1-p_{AR}}\right) = \log([A]) - \log K_A \qquad \text{eq.13}$$

If we plot log (p_{AR}/(1 - p_{AR})) on our 'y' axis against log[A] on the x-axis, this should follow the equation of a straight line of unit slope. Such a graph is described as a ***Hill plot***, again after A.V. Hill

who was the first to employ it, and is often used when p_{AR} is measured directly with a radiolabelled ligand. In practice, the slope of the line is not always unity (as discussed below) and is referred to as the ***Hill coefficient*** (n_H) or ***Hill slope***.

The relationship between receptor occupancy and tissue response

A.J. Clark and A.V. Hill were the pioneers in modelling the complete concentration-response relationship. In the absence at that time of any means of obtaining direct evidence on the point, they explored the consequences of assuming (1) that the law of mass action applies, so that Equation 11 (derived above) holds, and (2) that the response of the tissue is linearly related to receptor occupancy. Clark went further and made the *tentative* assumption that the relationship might be one of direct proportionality (though he was well aware that this was almost certainly an oversimplification, as we now know it often is).

Should there be direct proportionality, and if the response of a tissue (expressed as a percentage of the maximum response attainable with a large concentration of the drug) is denoted by *y,* the relationship between occupancy[1] and response becomes:

$$\frac{y}{y_{max}} = p_{AR} \qquad \text{eq.14}$$

Combining this with Equation 11 provides

$$\frac{y}{y_{max}} = \frac{[A]}{K_A+[A]} \qquad \text{eq.15}$$

Equation 15 can be rearranged to

$$\frac{y}{y_{max}-y} = \frac{[A]}{K_A} \qquad \text{eq.16}$$

Taking logs we get,

$$\log\left(\frac{y}{y_{max}-y}\right) = \log[A] - \log K_A \qquad \text{eq.17}$$

This can be tested by measuring a series of responses (y) to different concentrations of A, and then plotting log (y/(100 - y)) (setting y_{max} to 100) against log [A] (the Hill plot). If Equation 17 holds, a straight line with a slope of 1 should be obtained. A.J. Clark was the first to test this using the responses of isolated tissues, and Figure 2 shows some of his results. Equation 17 provides a good fit to the experimental values, and the slopes of the Hill plots in *B* are close to unity. While these findings are in keeping with the simple model that has been outlined, they do not amount to proof that it is correct. Indeed, later studies with a wide range of tissues have shown that some concentration-response relationships cannot be fitted by equation 15. In particular, the Hill coefficient is generally, if not always, greater than unity for responses mediated by ligand-gated ion channels[2]. Also, it is now known that in many tissues the maximal response (for example, contraction of intestinal smooth muscle) may occur when an agonist such as acetylcholine occupies less than a tenth of the available receptors. By the same token, when an agonist is applied at the concentration needed to give a half maximal response (termed the $[A]_{50}$ or EC_{50}), receptor occupancy may be as little as 1% in some tissues, rather than the 50% to be expected were the response to be directly proportional to occupancy.

[1]Note that no distinction is made here between *occupied* and *activated* receptors: it is tacitly assumed that all the receptors occupied by agonist molecules are in an active state, hence contributing to the initiation of the observed response. As we shall see later, this is an oversimplification.

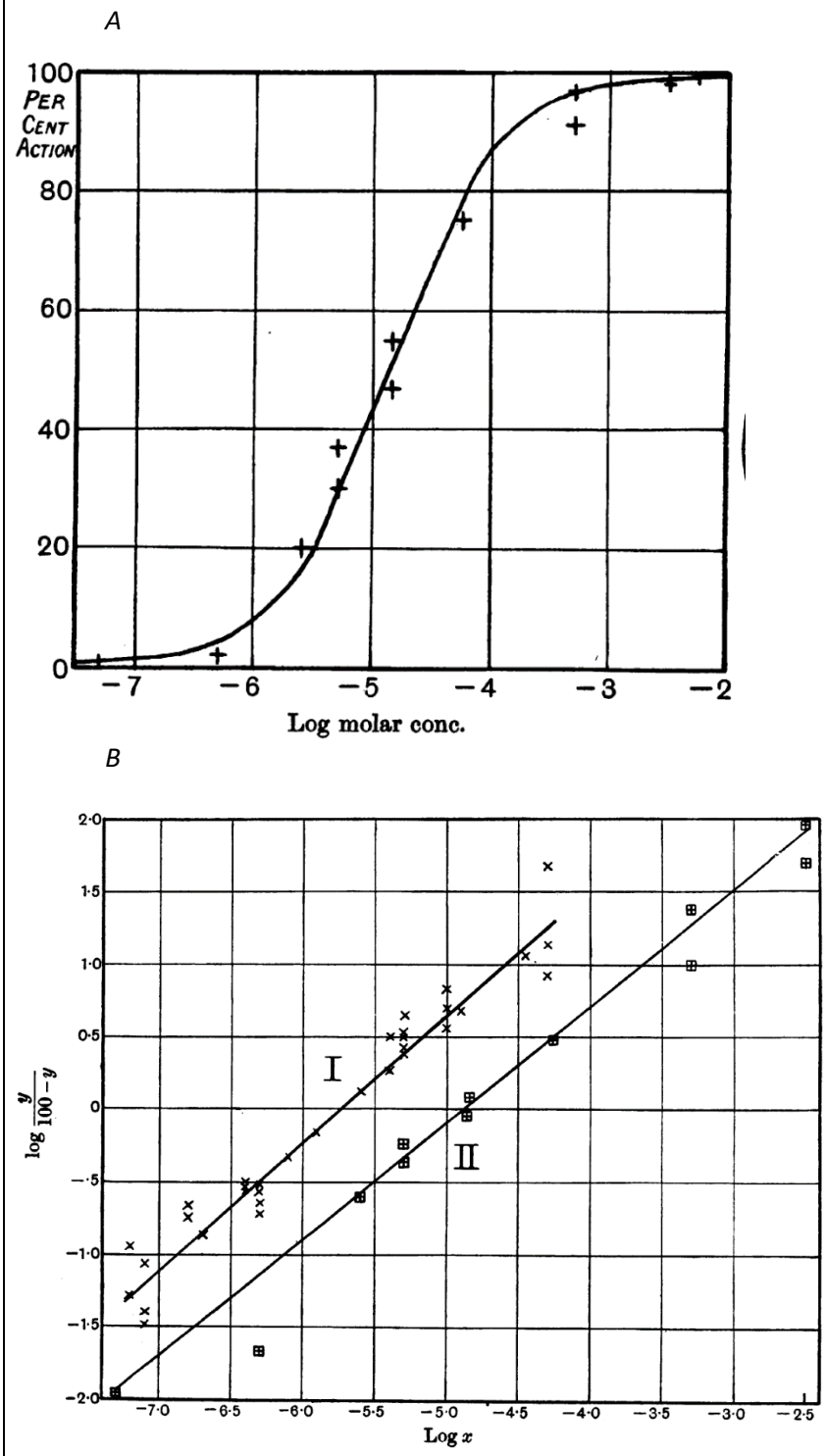


Figure 2. *A* Concentration-response relationship for the action of acetylcholine in causing contraction of the frog rectus abdominis muscle. The curve has been drawn using Equation 15.

B Hill plots for the action of acetylcholine on frog ventricle (curve I) and rectus abdominis (curve II).

From A.J. Clark, *J. Physiol*. **61**, 530 - 546, 1926.

Pharmacologists have therefore had to abandon (sometimes rather reluctantly and belatedly) their attempts to explain the shapes of the dose-response curves of complex tissues in terms of the simple models first explored by Clark and by Hill. Nevertheless, as Clark's work showed, the relationship between the concentration of an agonist and the response of a tissue commonly has the same

general form seen in Figure 1. In keeping with this, concentration-response curves can often be fitted *empirically* by the simple expression:

$$y = y_{\max}\frac{[\mathrm{A}]^{n_H}}{[\mathrm{A}]_{50}^{n_H}+[\mathrm{A}]^{n_H}} \qquad \text{eq.18}$$

This is usually described as the ***Hill equation***. Here n_H is the ***Hill coefficient*** and y and y_{max} are respectively the observed response and the maximum response to a large concentration of agonist. $[A]_{50}$ (or the equivalent term EC_{50}) is the molar concentration of A at which y is half maximal. Because it is a constant for a given concentration-response relationship, it is sometimes denoted by *K*. While this is algebraically neater (and moreover was the symbol used by Hill), it should be remembered that *K* in this context does not necessarily correspond to an equilibrium constant. Employing $[A]_{50}$ in equation 18 serves to remind us that the relationship between [A] and response is here being *described* rather than *explained* in terms of a model of receptor action.

Finally: To return to our question "will it help to take two ibuprofen?". Ibuprofen is a drug that inhibits two similar enzymes in the body - cyclo-oxygenase-1 (COX-1) and cyclo-oxygenase-2 (COX-2). Suppose we treat the inhibition of these enzymes as a simple equilibrium reaction of the following kind:

$$\mathrm{P} + \mathrm{E} \underset{}{\overset{K_\mathrm{A}}{\rightleftharpoons}} \mathrm{PE}$$

Where P represents our painkiller (ibuprofen) and E represents the COX-2 enzyme. Whether or not there is any benefit from taking two, rather than one tablet, depends on where we are on the dose-response curve. To confirm this point why not try the following calculation. Assuming that the proportion of inhibited enzyme p_{PE} is given by:

$$p_{\mathrm{PE}} = \frac{[P]}{K_\mathrm{A}+[P]}$$

and that the K_A (dissociation constant) for ibuprofen binding is approximately 10 µM. What would be the effect of doubling the dose, assuming that a single tablet raised the free concentration of ibuprofen to (i) 5 µM or (ii) (probably more realistically) 24 µM. Hopefully you will see that it makes less difference than you might expect to double the dose. However, this is by no means a complete answer because we need to factor in all the issues mentioned at the start of these notes. We also need to discuss 'spare receptors' ….see your next lecture.

APPENDIX 1. Equilibrium, dissociation and affinity constants

Confusingly, all these terms are in current use to express the position of the equilibrium between a drug and its receptors. The choice arises because the ratio of the rate constants k_{-1} and k_{+1} can be expressed either way up. In these notes we take K_A to be k_{-1} / k_{+1} , and it is then strictly a *dissociation equilibrium constant*, which is often abbreviated to either *dissociation constant* or *equilibrium constant*. The inverse ratio, k_{+1} / k_{-1} , gives the *association equilibrium constant* which is always abbreviated to *affinity constant*.

One way to reduce the risk of confusion is to express drug concentrations in terms of K_A. This 'normalised' concentration, $¢_A$, is defined as $[A]/K_A$. We can therefore write the Hill-Langmuir equation in three different though equivalent ways:

$$p_{AR} = \frac{[A]}{K_A + [A]} = \frac{K_A'[A]}{1 + K_A'[A]} = \frac{¢_A}{1 + ¢_A}$$

where the terms are as follows:

	abbreviation	***dimension***
dissociation equilibrium constant	K_A	M
affinity constant	K'_A	M^{-1}
normalised concentration	$¢_A$	dimensionless

Appendix 2. Step-by-step derivation of the Hill-Langmuir equation

Starting point:

$$p_R + p_{AR} = 1 \text{ .....(1)}$$

$$[A]\, p_R = K_A\ p_{AR} \text{ .....(2)}$$

From (2), $p_R = \frac{K_A}{[A]} . p_{AR}$.....(3)

(Remember, if $ax = by$ then $x = \frac{b}{a} y$)

Next, use eqn. (3) to replace p_R in eqn (1)
(because we wish to find p_{AR})

$$\frac{K_A}{[A]} . p_{AR} + p_{AR} = 1$$

$$p_{AR}\left(\frac{K_A}{[A]} + 1\right) = 1$$ Remember, $ax + x = x\,(a+1)$

$$p_{AR}\left(\frac{K_A + [A]}{[A]}\right) = 1$$ Remember, $\frac{s}{t} + 1 = \frac{s+t}{t}$

$$\boxed{p_{AR} = \frac{[A]}{K_A + [A]}}$$ Remember if $x\frac{u}{v} = 1$ then $x = \frac{v}{u}$

Rearrangements of the Hill-Langmuir equation.

Cross-multiplying

$p_{AR}\, K_A + p_{AR} . [A] = [A]$

$\therefore\ p_{AR} . K_A = [A]\,(1 - p_{AR})$

Remember, $y = \frac{x}{a+x}$

is the same as $\frac{y}{1} = \frac{x}{a+x}$,

ready for cross-multiplication.

Cross multiplication:

if $\frac{a}{b} = \frac{c}{d}$, then a x $d = c$ x b

$$\boxed{\frac{p_{AR}}{1 - p_{AR}} = \frac{[A]}{K_A}}$$

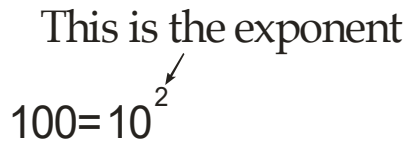
Taking logs,

$$\log\left(\frac{p_{AR}}{1-p_{AR}}\right) = \log[A] - \log K_A$$

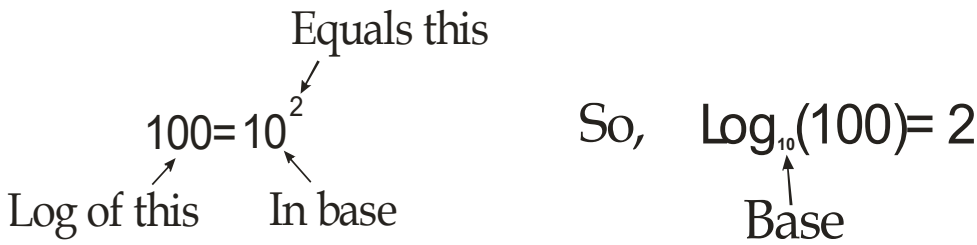
Remember, log (a/b) = log a - log b

Appendix 3: Logarithms and exponentials

You should be familiar with the idea of an exponent (also called an index or power). You will have come across it in a situation something like this:



Logarithms are another way of looking at the exponent, index or power. A logarithm has a base, and the logarithm of a number 'x' is simply that power to which that base must be raised in order to obtain 'x'. What this means is most easily seen by example; the logarithm of 100 (in log base 10) is 2. This is true because 10 (the base) raised to the power 2 is equal to 100. Similarly, the logarithm of 1000 (in base 10) is 3 and the logarithm of 10,000 (again in base 10) is 4. The following diagram may help if further explanation is needed:



You will have noticed that the logarithm of a very large number is, relatively speaking, a much smaller number. This makes logarithms particularly useful when we want to deal with a very wide range of numbers. Consider, for example, the following data set:

Conc. (nM)	Log(Conc.)	Response
1	0	0.01
10	1	0.1
100	2	0.5
1000	3	0.91
10000	4	0.99

If this data is plotted on a linear scale, then the final data point dominates the *x*-axis and all the other data points are 'squashed' near the start of the axis. On a logarithmic scale however, the same data can be displayed much more clearly. These points are illustrated in the plots below. This property of logarithms makes them ideal for many uses in biology.

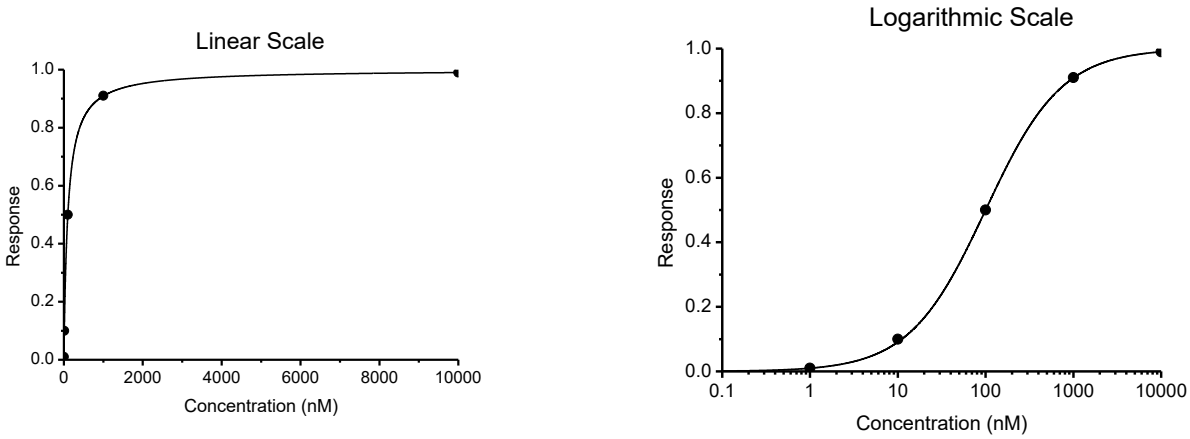


Figure 5. On a linear scale (left hand graph), a wide range of data points can be difficult to visualise because the axis range is dominated by the largest value. On a logarithmic scale the data are more evenly spaced and the concentration required, for example, to obtain 50% of the maximum response (100 nM), is easily seen.

Rules of Logarithms

Three important rules about logarithms are:

Rule 1 $Log(x^a) = a \times Log(x)$

Rule 2 (Proof first). Let $Log_a(b) = x$

and $Log_a(c) = y$

Then it must be true that $a^x = b$ and $a^y = c$

Therefore $a^x \times a^y = bc$

And $a^{x+y} = bc$

So $Log(bc) = x + y$

Rule 2 $Log(bc) = Log(b) + Log(c)$

Similarly one can prove that:

Rule 3 $Log\left({}^{b}/_{c}\right) = Log(b) - Log(c)$